

Exaai — MVP Review

Detailed Issues, Reasoning & Color-Coded Development To-Do List

This version includes the detailed reasoning behind the review remarks, followed by the consolidated development to-do list. Priority levels are represented with consistent color codes throughout the document.

PRIORITY COLOR CODE

Priority	Color	Meaning
CRITICAL		Fundamental to the product vision, correctness, or core exam-generation requirements.
MUST		Required for the final product; may be deferred from the MVP where explicitly stated.
IMPORTANT		Important final-version requirement, but not necessarily required to validate the MVP.
OPTIONAL		Useful enhancement that can be implemented later without affecting the core product.

PART I — DETAILED ISSUES & REASONING

1. General Workflow

Issue: The current processes are separated into individual tabs. The preferred final experience is one continuous exam workspace containing the text and subsequent sections.

Reasoning: Teachers are accustomed to scanning the text and moving back and forth between it and the questions on the same exam sheet. Keeping the workflow together makes Exaai feel natural and intuitive, reduces friction, and more closely reproduces the traditional exam-making process. A familiar workflow should also make demonstrations and marketing easier.

Decision: Final product: unified editable exam workspace. MVP: separate tabs are acceptable if restructuring would materially delay MVP testing.

2. Parameters — Topic Selection

Issue: The current topic suggestion list is too limited and does not provide enough meaningful choice.

Reasoning: Topic selection is one of the points where the teacher is expected to spend the most time. A small list makes the product restrictive. The topic area therefore needs substantially more space and at least 20+ relevant options. Custom-topic suggestions should behave more like search suggestions, drawing from authentic/adapted sources rather than randomly invented AI topics.

Decision: Expand the topic pool, improve its presentation, and make custom-topic suggestions source-informed. The exact visual presentation will be specified separately.

3. Parameters — Ministry Units

Issue: Some units currently do not exactly match the Ministry syllabus.

Reasoning: This does not need to block a hypothetical-data MVP whose purpose is testing the workflow. However, accurate Ministry mapping is necessary in the final product because units influence topic selection and appropriate exam content.

Decision: Final: align units with the Ministry syllabus. MVP: current approximation may remain.

4. Parameters — Teacher Keywords

Issue: There is no dedicated field for keywords or concepts the teacher wants included in the text.

Reasoning: Teachers may want the exam to reinforce vocabulary or concepts already taught. A keyword field gives meaningful control without requiring the teacher to write the text manually.

Decision: Add a teacher-keyword field in the final version; MVP implementation is optional.

5. Text Generation — Modification

Issue: Simplifying, hardening, or otherwise modifying a text creates a new alternative rather than modifying the current text.

Reasoning: The teacher is normally asking to edit the selected text, not create another candidate. The natural workflow is to update the current text and provide Undo.

Decision: Current text → Modify → Modified current text → Undo if needed. Implement in MVP if possible and retain in final.

6. Text Generation — Authentic Sources

Issue: The MVP appears to generate texts directly with AI rather than extracting/adapting material from real online sources.

Reasoning: Authentic/adapted source material is central to the intended product. The teacher needs an exam text that can be presented with a real source, not simply an AI-invented paragraph. The current limitation may be caused by the MVP AI being basic or disconnected from online data, but this should not define the final product.

Decision: Critical for the final version and critical for the MVP if technically feasible. The precise definition of acceptable authentic/adapted sources will be provided separately.

7. Text Generation — In-text Assistance

Issue: The AI currently does not offer contextual help within the text, such as summarising, synonyms or rephrasing.

Reasoning: These capabilities are useful but do not test the core exam-building workflow. They should not compete with higher-priority work during the MVP.

Decision: Optional/future functionality.

8. Part One — Answer Key

Issue: Answer-key generation was added and is a useful feature.

Reasoning: Since the purpose of Exaai is to reduce teacher workload, generating the answer key completes an important part of the exam-building process.

Decision: Keep and refine answer-key generation.

9. Part One — 2017 Ministry Guide

Issue: The generated Reading Comprehension tasks do not sufficiently comply with the official 2017 Ministry Guide. Problems include True/False statement counts, question counts, task variety/number, and stream-specific requirements.

Reasoning: Exaai cannot rely on the AI's generic idea of a good comprehension exercise. The official guide determines the expected structure, and the selected stream can change that structure. These requirements therefore need to govern generation directly.

Decision: Critical for both MVP and final. The 2017 guide will be provided to the developer as the reference.

10. Part One — Change Task

Issue: Change Task currently produces essentially the same task with different statements or questions.

Reasoning: The purpose is to give the teacher a different legitimate way of assessing the text, not superficial variation. Alternatives should represent different task types permitted by the Ministry guide and appropriate to the text, stream and marks.

Decision: Change Task must offer genuinely different task types. Maintain the alternative-task popup concept.

11. Part B — Morphology

Issue: Some morphology tasks, such as noun/verb/adjective transformations, require a table that is currently missing.

Reasoning: The visual structure is part of the task. A table makes the relationship between word forms clear and better matches the expected exam format.

Decision: Add dynamic table structures where required.

12. Part B — Ministry Guide

Issue: Part B is generally better than earlier sections and its order/task types are acceptable, but it still does not fully follow the Ministry guide.

Reasoning: High-level similarity is not enough if task counts, marks, ordering or other requirements are wrong. Part B needs the same rule-based verification as Part One.

Decision: Bring Part B into full Ministry-guide compliance while preserving its generally acceptable current structure.

13. Written Expression — Ministry Guide

Issue: Written Expression also needs to follow the Ministry guide more closely.

Reasoning: It should not simply contain two generic AI writing prompts. The guided/free distinction, situation format, marks and exam configuration need to reflect the expected exam structure.

Decision: Treat Ministry-guide compliance as an important final-version requirement.

PART II — COLOR-CODED DEVELOPMENT TO-DO LIST

The same priority colors are used here so the developer can immediately distinguish the implementation hierarchy.

1. GENERAL WORKFLOW

CRITICAL — Final / MVP optional

- Unify the exam-building workflow into one workspace/page.
 - Text, Part One, Part B/Text Exploration, and Written Expression should eventually appear as sections of the same exam.
 - Allow natural movement between text and questions.
 - Maintain editing throughout the process.
- MVP:** Separate tabs may remain if unification would significantly delay testing.

2. PARAMETERS

CRITICAL — MVP + Final — Topic Selection

- Significantly expand topic suggestions.
- Provide at least 20+ relevant topics for the selected unit.
- Match level, grade, stream and unit.
- Give the topic area considerably more visual space.
- Replace the current simple list with a more suitable interface; exact design to be discussed separately.
- Base topics on authentic/adapted source material.
- For custom keywords, suggest related topics based on relevant authentic/adapted sources.
- Make suggestions behave more like search suggestions than random AI ideas.

MUST — Final / MVP optional — Ministry Units

- Align units with the actual Ministry syllabus.
- Ensure correct mapping to level, grade and stream.

MUST — Final / MVP optional — Teacher Keywords

- Add a field for keywords/concepts that should appear in the generated text.
- Attempt to incorporate them naturally.

3. TEXT GENERATION

MUST — MVP if possible + Final — Text Modification

- Modify the current text rather than creating another alternative.
 - Make the modified version the active text.
 - Add Undo.
 - Do not add every modification as another alternative.
- Flow:** Current Text → Modify → Modified Current Text → Undo if needed.

CRITICAL — MVP if possible + Final — Authentic/Adapted Sources

- Use authentic/adapted online sources rather than entirely AI-created texts.
- Identify and automatically cite the source.
- Select source material according to topic and learner level.
- Adapt to learner level while maintaining source authenticity.
- Apply the separately supplied source criteria.

OPTIONAL — Future — In-text AI Assistance

- Simplify, make harder, rephrase, synonyms, vocabulary assistance, summarize selected text.

4. PART ONE — READING COMPREHENSION

MUST — MVP + Final — Answer Key

- Keep answer-key generation.
- Ensure answers correspond to every task.
- Eventually support all task types.

CRITICAL — MVP + Final — Ministry Guide

- Use the 2017 Ministry Guide as the generation reference.
- Correct True/False statement counts.
- Correct question counts.
- Correct task variety and number of tasks.
- Respect selected stream.
- Correct marks and task distribution.
- Use the official guide once provided.

CRITICAL — MVP + Final — Change Task

- Offer genuinely different task types.
- Ensure alternatives are Ministry-compliant.
- Match selected stream and text.
- Respect marks.
- Maintain/improve the alternative-task popup.

5. PART B — TEXT EXPLORATION

MUST — MVP + Final — Morphology

- Add appropriate tables for morphology tasks.
- Support noun/verb/adjective transformations and similar formats.
- Adapt table structure to the generated task.

CRITICAL — MVP + Final — Ministry Guide

- Verify task types, order, number of tasks, marks, stream alignment and intended language skills against the official guide.

6. WRITTEN EXPRESSION

IMPORTANT — Final — Ministry Guide

- Verify number/type of writing topics.
- Verify marks.
- Preserve guided/free distinction.
- Match selected exam/stream.
- Ensure situations and instructions follow the appropriate exam format.

7. MVP PRIORITY ORDER

1. ■ Ministry-compliant Part One.
2. ■ Proper Change Task functionality.
3. ■ Better topic selection with 20+ relevant options.
4. ■ Authentic/adapted text generation if feasible.

5. ■ Current-text modification + Undo.
6. ■ Part B Ministry compliance.
7. ■ Morphology tables.
8. ■ Answer key refinement.
9. ■ Teacher keywords.
10. ■ Unified workspace, if it does not significantly delay MVP testing.

8. FINAL-VERSION PRIORITIES

- Unified exam workspace.
- Accurate Ministry syllabus/unit structure.
- Comprehensive source-backed topic discovery.
- Authentic/adapted source retrieval.
- Teacher keyword control.
- Full Ministry-compliant generation across all sections.
- Proper task replacement system.
- Correct tables and specialized task layouts.
- Written Expression compliance.
- Contextual AI assistance inside the text.